Year 10 Mathematics

Sets 1

9.1 Set Notation

Set Notation

STARTER · 5 MIN

Date: _____ **Title:** Set Notation

LEARNING OBJECTIVES

- Describe sets using lists and words
- Use the symbols $\in$, $\notin$, n(A), $\mathcal{E}$ and $\varnothing$
- Understand subsets

STARTER *Answer in your book. No calculators.*

1. List the factors of 12
2. List the prime numbers less than 15
3. List the multiples of 4 between 10 and 30
4. List the square numbers less than 50

Starter — Answers

STARTER

Tick or correct in green pen

1. 1, 2, 3, 4, 6, 12

2. 2, 3, 5, 7, 11, 13

3. 12, 16, 20, 24, 28

4. 1, 4, 9, 16, 25, 36, 49

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can describe sets using lists and words	☐
SILVER	I can use the symbols $\in$, $\notin$, n(A), $\mathcal{E}$ and $\varnothing$	☐
GOLD	I can understand subsets	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Set Notation

A set is a collection of objects, called elements, written inside curly brackets { }.

Elements

$\in$ means 'is an element of'. $\notin$ means 'is not'.

$3 \in \{1,\ 2,\ 3\},\ 5 \notin \{1,\ 2,\ 3\}$

Number of elements

n(A) is how many elements are in A.

A = {2, 4, 6}, n(A) = 3

Universal and empty sets

$\mathscr{E}$ contains everything. $\varnothing$ is the empty set.

$\varnothing = \{\ \}$

Subsets

$A \subset B$ means every element of A is in B.

$\{2,\ 4\} \subset \{1,\ 2,\ 3,\ 4\}$

Examples 1 and 2 — Listing Elements

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

A = {even numbers between 1 and 11}

List the elements of A and find n(A).

Example 2

B = {vowels in the word MATHS}

List the elements of B.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

A = {2, 4, 6, 8, 10}

n(A) **= 5**

Tip: n(A) counts the elements.

Example 2

Vowels in MATHS: A

B = {A}

Tip: each element is listed only once.

Task 1

PRACTICE · 9 MIN

BRONZE

List the elements of each set

1. {odd numbers between 2 and 12}
2. {factors of 18}
3. {prime numbers between 10 and 30}
4. {letters in the word SETS}
5. {multiples of 7 less than 40}

★ Find n(A) for A = {square numbers less than 100}

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. **{3, 5, 7, 9, 11}**

2. **{1, 2, 3, 6, 9, 18}**

3. **{11, 13, 17, 19, 23, 29}**

4. **{S, E, T}**

5. **{7, 14, 21, 28, 35}**

★ A = {1, 4, 9, 16, 25, 36, 49, 64, 81}, n(A) = **9**

Examples 3 and 4 — Using Set Symbols

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

$A = \{1, 3, 5, 7, 9\}$

Is $6 \in A$ true or false?

Example 4

$P = \{2, 3, 5, 7\}$ and $Q = \{1, 2, 3, 4, 5, 6, 7\}$

Is $P \subset Q$? Explain.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

6 is not in A

$6 \in A$ is false

Tip: $\in$ means 'is in'.

Example 4

2, 3, 5 and 7 are all in Q

Yes, $P \subset Q$

Tip: check every element.

Task 2

PRACTICE · 9 MIN

SILVER

A = {2, 4, 6, 8}. B = {1, 2, 3, 4}. Answer each question.

1. Is $6 \in A$ true or false?

2. Is $3 \notin B$ true or false?

3. Find n(B).

4. Is $\{2,\ 4\} \subset A$? Explain.

5. Write down an element of A that is also in B.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 6 is in A, **True**
2. 3 is in B, **False**
3. n(B) = **4**
4. 2 and 4 are in A, **Yes**
5. **2 or 4**

Examples 5 and 6 — Describing Sets

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5

(1 mark)

Describe in words the set

{5, 10, 15, 20, 25}

Example 6

(2 marks)

$\mathscr{E}$ = {integers from 1 to 20}

C = {x : x is a square number}

List the elements of C.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Each number is a multiple of 5

{multiples of 5 from 5 to 25}

Tip: look for the pattern.

Example 6

Square numbers up to 20

C = {1, 4, 9, 16}

Tip: only use elements in $\mathcal{E}$.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Describe in words: {1, 3, 5, 7, 9} (1 mark)

2. $\mathscr{E} = \{1,\ 2,\ \ldots,\ 15\}$. $D = \{x : x \text{ is prime}\}$. List D. (2 marks)

3. $\mathscr{E} = \{1,\ 2,\ \ldots,\ 12\}$. $E = \{x : 3x > 20\}$. List E. (2 marks)

4. Write down all the subsets of {a, b}. (2 marks)

5. F = {x : x is an even prime number}. Find n(F). (1 mark)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **{odd numbers less than 10}**

2. D = **{2, 3, 5, 7, 11, 13}**

3. 3x > 20, x > 6.67, E = **{7, 8, 9, 10, 11, 12}**

4. ∅, {a}, {b}, **{a, b}**

5. F = {2}, n(F) = **1**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE List {factors of 20}

SILVER $A = \{3,\ 6,\ 9,\ 12\}$. Find n(A) and say if $8 \in A$.

GOLD $\mathscr{E} = \{1,\ 2,\ \ldots,\ 10\}$. List $\{x : x^2 < 30\}$.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **{1, 2, 4, 5, 10, 20}**

SILVER $n(A) = 4, 8 \notin A$

GOLD **{1, 2, 3, 4, 5}**

Exam Practice — Listing Elements

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

A = {factors of 20}

(a) List the elements of A. (1)

.....

(b) Find n(A). (1)

n(A) = ...

Question 2

(2 marks)

B = {letters in the word ELEMENT}

(a) List the elements of B. (1)

.....

(b) Find n(B). (1)

n(B) = ...

Exam Practice — Using Set Symbols

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(3 marks)

$A = \{2, 4, 6, 8, 10\}$

$B = \{4, 8\}$

(a) Is $6 \in A$ true or false? (1)

.....

(b) Is $4 \notin B$ true or false? (1)

.....

(c) Is $B \subset A$? Give a reason. (1)

.....

Question 4

(2 marks)

$\mathcal{E} = \{1, 2, 3, \ldots, 10\}$

$A = \{1, 2, 3, 4, 5\}$

(a) Find x where $x \in \mathcal{E}$ and $x \notin A$. (1)

x = ...

(b) Write down a subset of A with exactly two elements. (1)

.....

Exam Practice — Set-Builder Notation

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

$\mathcal{E} = \{1, 2, 3, \ldots, 20\}$

$D = \{x : 3x + 1 < 20\}$

(a) List the elements of D. (2)

D = ...

(b) Is $7 \in D$? Give a reason. (1)

.....

Question 6

(2 marks)

$S = \{1, 4, 9, 16, 25\}$

$G = \{x : x \text{ is an odd multiple of } 2\}$

(a) Describe the set S in words. (1)

.....

(b) Find n(G). (1)

n(G) = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) {1, 2, 4, 5, 10, 20} B1

(b) n(A) = 6 B1

Tip: find factors in pairs: 1 × 20, 2 × 10, 4 × 5.

Question 2

(a) {E, L, M, N, T} B1

(b) n(B) = 5 B1

Tip: list each element only once.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) True B1

(b) False B1

(c) Yes, 4 and 8 are in A B1

Tip: ∈ means 'is in', ∉ means 'is not in'.

Question 4

(a) any of 6, 7, 8, 9, 10 B1

(b) e.g. {1, 2} B1

Tip: any two elements of A are correct.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $3x < 19$, so $x < 6.33…$ — M1

D = {1, 2, 3, 4, 5, 6} — A1

(b) No: 3 × 7 + 1 = 22 — B1

Tip: only use elements in $\mathcal{E}$.

Question 6

(a) {square numbers up to 25} — B1

(b) G = ø, so n(G) = 0 — B1

Tip: ø is the empty set, with no elements.